ENVIRONMENTAL SCIENCE (MC-ES201)

Course Code: MC-ES201 | Mandatory Non-Credit Course | 3 Credits Equiv.

Maulana Abul Kalam Azad University of Technology - Official Curriculum Scheme

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Unit 1: Ecosystems & Biodiversity Conservation

Ecosystem concepts, biotic and abiotic components, energy flow, food chains and ecological pyramids. Biodiversity hotspots in India, threats to biodiversity, in-situ and ex-situ conservation.

Unit 2: Environmental Pollution Air, Water, Soil, Noise

Classification of pollutants, greenhouse effect, photochemical smog, wastewater treatment processes, soil degradation, noise pollution standards, solid waste and e-waste management.

Unit 3: Renewable & Non-Renewable Natural Resources

Forest resources and deforestation, water resources and conflict, dams, solar and wind energy harvesting, biomass, geothermal energy, mineral extraction impacts.

Unit 4: Environmental Policies, Climate Change & Treaties

Climate change science, ozone layer depletion, acid rain, Environmental Protection Act 1986, Air and Water pollution acts, Montreal Protocol, Kyoto Protocol, Paris Agreement.

Unit 5: Fieldwork, Waste Management & Sustainability

Case studies of local polluted sites, carbon footprints, circular economy principles, sustainable development goals SDGs, disaster mitigation techniques.